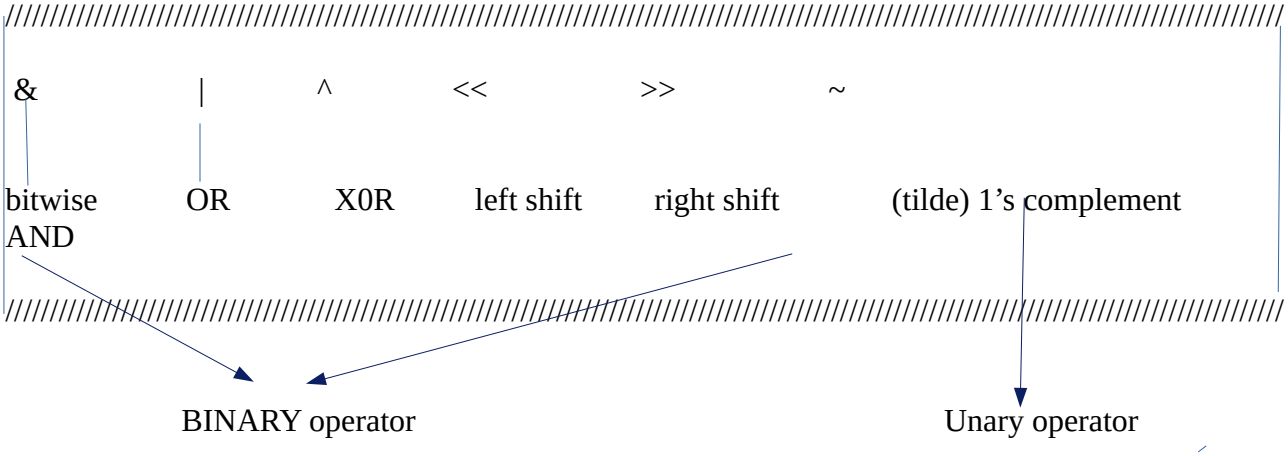
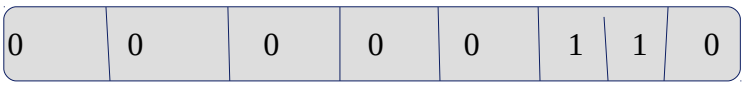
Bitwise operators



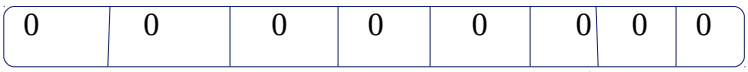
Bitwise operator not applicable on Real data.

```
 intmain()
{
   int a=6,b=1,c=0;
   c=a&b;
   printf("%d", c);
  c=a|| b;printf("%d  ", c);
 c=a^b; printf("%d ",c);
}
```



a=6

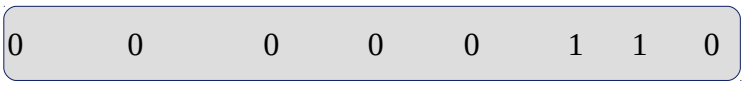
b=1



c= a & b

result =0

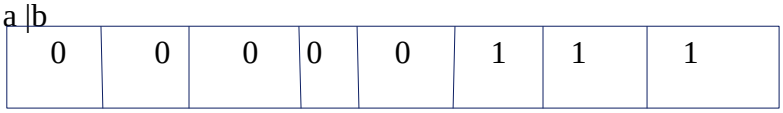
Q: c= a | b;



a=6

b=1

c= a |b

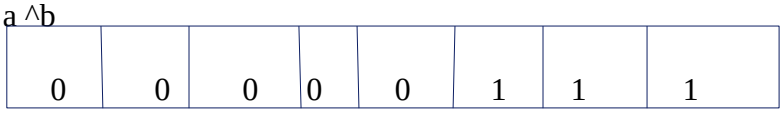


c= 7

//

c= a^ b

c= a ^b



c= 7

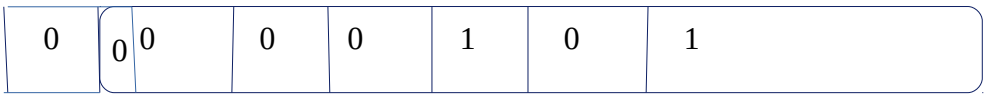
Bitwise not (~) /tilde

1s complement =2's complement -1

2's complement =1 's complement +1;
1's complement =2's complement -1;

1's complement= - data -1 → shortcut

Q: find ~5



~ 5= ~ 0 0 0 0 0 1 0 1

RESULT= 1 1 1 1 1 0 1 0 = -6

Take 2's complement of result to verify

```
0  0  0  0  0  1  0  1
+                  1
-----------------------
0  0  0  0  0  1  1  0
```

//////this is binary represenation of 6

hence 1 1 1 1 1 0 1 0 =-6 verified

//

using formula, 1s complemnet= 2,s complement -1

= - data -1

~ 5 = - 5 -1
= -6

//
~ -8 (negative values are stored in 2's complement form)

0 0 0 0 1 0 0 0 =8

To get - 8 ,take 2's complement

2's complement = 1's complement + 1

```
1's com=  1  1  1  1  0  1  1  1
+                             1
-----------------------------------
          1  1  1  1  1  0  0  0
```

= -8

Now ~(-8)

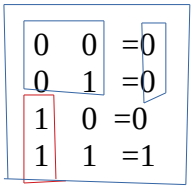
~ 1 1 1 1 1 0 0 0

result= 0 0 0 0 0 1 1 1

```
int a=20,b=0;b=~(a+30);
printf("%d",  b);
printf("%d",~-456);
printf("%d",~(100+200));
printf("%d",~5 +~6);
```

//

AND



0 & with any data= 0;

---

0 0 0 0 0 0 0 0 0
&
x 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0////////////0

.....

.....

0 0 0 0 0 0 0 0 0
&
x 1 1 1 1 1 1 1 1
0 0 0 0 0 0 0 0 0
.....////////////////////0

.....

-1 1 1 1 1 1 1 1 1

&

x b b b b b b b b
=x ///////depends on data bit

.....

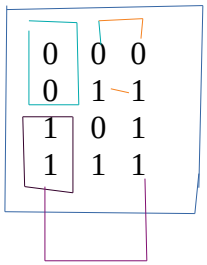
```
int main()

{

 printf("%d  ",20 &0); //0
 printf("%d   ", 0 & 0);//0
 printf("%d  ",0 &-1);//0
 printf("%d  ",30&-1);//30
 printf("%d  ",1200 & -1);//1200
}
```

//

0R




```
-------------------------------------
0      0 0   0   0 0   0 0 0
|
x     1    1    1  1   1   1   1  1
```

=X(depends on bit)

.....

```
-----------------------------------------
 -1     1  1  1   1     1   1  1   1

|

 x      b  b   b    b   b   b   b   b
```

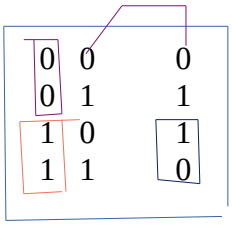
```
     1   1   1    1    1    1   1   1   ======  -1
```

.....

50 / 0 =50
50 /-1 =-1
12345/0 =12345
12345/-1 =-1

//

XOR



//
0 0 0 0 0 0 0 0 0
^
data b b b b b b b b

---

data b b b b b b b b

//

//
-1 1 1 1 1 1 1 1 1 1

^
data b b b b b b b b b

.....
(~data)

```
main()
{
printf("%d  ",50  ^ 0); //50
printf("%d  ",1234 ^ 0) /1234
printf("%d  ",  50 ^  -1);  //complemnet of data (~ 50) =-51;
}
```